Integrative genetic, transcriptomic and single-cell analysis of IL6R in colorectal cancer links IL6R to immune and myeloid tumour microenvironments despite limited circulating-protein causal evidence

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Abstract

Background: IL-6/IL6R signalling is implicated in inflammatory regulation and tumour-microenvironment biology, but whether circulating IL6R has a detectable causal relationship with colorectal cancer (CRC) risk remains uncertain. We integrated genetic epidemiology, tumour transcriptomics, clinical endpoint reconstruction, single-cell RNA sequencing and network-support analyses to clarify how IL6R should be interpreted in CRC.

Methods: Two-sample Mendelian randomization (MR) evaluated genetically proxied circulating IL6R protein levels and CRC risk across a primary FinnGen endpoint and two replication CRC GWAS endpoints. Colocalization was performed at the IL6R locus. TCGA-COAD/READ analyses assessed IL6R expression, reconstructed overall survival (OS) and progression-free interval (PFI), and immune-infiltration correlations. GSE146771 single-cell RNA-seq data were used to localize IL6-axis expression and CellChat-supported ligand-receptor communication. IL6R-centred co-expression, enrichment and drug-target support analyses were used to contextualize immune, myeloid and inflammatory programmes.

Results: MR did not support a significant causal effect of genetically proxied circulating IL6R protein levels on CRC risk. Colocalization at the IL6R locus also did not support a strong shared causal variant (PP.H4 = 0.048). TCGA analyses showed markedly lower IL6R expression in tumours than adjacent normal tissues but did not support IL6R as a robust OS/PFI prognostic marker after patient-level survival reconstruction. External GEO validation extracted IL6R expression from three independent CRC cohorts (GSE17536, GSE17537 and GSE14333; n = 522) and did not identify significant univariable survival associations in GSE17536 or GSE17537. In contrast, immune-infiltration, single-cell and support-layer analyses consistently linked IL6R to immune-, myeloid- and inflammatory-associated tumour microenvironment features.

Conclusions: Genetic analyses did not support a strong causal or colocalized effect of circulating IL6R protein levels on CRC risk, and TCGA plus external GEO survival checks did not support IL6R as a robust standalone prognostic marker. Nevertheless, transcriptomic, single-cell and support-layer analyses consistently connected IL6R to an immune-, myeloid- and inflammatory-associated CRC tumour microenvironment.

Keywords: IL6R; interleukin-6; colorectal cancer; Mendelian randomization; colocalization; TCGA; single-cell RNA-seq; tumour microenvironment; myeloid inflammation; drug-target network

Background

Colorectal cancer (CRC) remains a major global malignancy and a leading contributor to cancer morbidity and mortality. Recent GLOBOCAN estimates indicate that CRC accounts for more than 1.9 million incident cases and approximately 904,000 deaths worldwide, underscoring the need to clarify molecular and immune-microenvironmental determinants of CRC biology [1]. Contemporary epidemiological updates also highlight persistent challenges related to late detection, recurrence, therapy resistance and early-onset disease [2]. These challenges make it important to distinguish molecular features that are plausible causal determinants of CRC risk from features that instead reflect tumour ecology, immune infiltration or tissue-context responses.

Inflammatory signalling is central to CRC biology because chronic intestinal inflammation, stromal remodelling and immune-cell recruitment can shape epithelial transformation and tumour progression. Interleukin-6 (IL-6) signals through IL6R and the signal-transducing co-receptor IL6ST/gp130, activating downstream JAK/STAT3 and related pathways involved in immune regulation, survival signalling and inflammatory feedback [3,4]. However, IL-6-axis biology is context-dependent: circulating cytokine or receptor levels, soluble receptor dynamics, membrane receptor expression and local cell-cell communication may represent different biological layers [5]. This creates an important interpretive challenge for IL6R in CRC.

Mendelian randomization (MR) can help evaluate whether genetically proxied molecular exposures are associated with disease risk under instrumental-variable assumptions [6,7]. Colocalization adds a complementary genetic layer by evaluating whether exposure and outcome signals in a locus are compatible with a shared causal variant rather than neighbouring signals in linkage disequilibrium [9]. Together, MR and colocalization can separate putative genetic-causality evidence from broader tissue-level or tumour-microenvironment associations.

Negative MR or colocalization evidence does not imply that a molecule is biologically irrelevant. Instead, it may indicate that the measured circulating exposure does not have a detectable germline causal effect on disease risk, whereas tissue-level expression or cell-specific signalling may still be associated with tumour states. TCGA-COAD/READ provides transcriptomic and clinical resources for tumour-normal expression, survival and immune-infiltration analyses [10-14]. Single-cell RNA sequencing further localizes expression and inferred ligand-receptor communication to tumour microenvironment compartments [15-17]. Finally, co-expression, pathway-enrichment and drug-target support layers can help determine whether IL6R-associated genes converge on immune, cytokine and myeloid programmes [18-21].

This study applied an integrated evidence-chain framework to IL6R in CRC. We first used MR and colocalization to test whether circulating IL6R protein levels were genetically associated with CRC risk and whether the IL6R locus supported shared causal variation. We then assessed IL6R expression, reconstructed OS/PFI endpoints and immune infiltration in TCGA-COAD/READ. Next, we used GSE146771 single-cell RNA-seq data to localize IL6-axis expression and communication. Finally, we constructed IL6R-centred co-expression, enrichment and drug-target support layers. The central aim was not to force a causal interpretation where genetic evidence is weak, but to position IL6R within an explicit evidence hierarchy: genetic-causality testing first, followed by transcriptomic and cellular-context interpretation.

Methods

Overall study design

This study was designed as an integrative multi-layer analysis to evaluate the relationship between IL6R-related biology and CRC. The overall workflow and evidence-chain logic are summarized in Figure 1. The analytical framework included two-sample MR, IL6R-locus colocalization, TCGA-COAD/READ transcriptomic and clinical analyses, GSE146771 single-cell RNA-seq analysis and support-layer network, enrichment and drug-target annotation. Because the MR and colocalization layers did not support a strong causal or shared-variant effect, downstream transcriptomic and network analyses were interpreted as biological-context support rather than as proof of causal therapeutic targeting.

Data sources and analytical layers were organized as follows. Genetic exposure analyses used circulating IL6R protein pQTL instruments to test genetic causal evidence. CRC outcome analyses used FinnGen CRC plus IEU and EBI/GWAS Catalog CRC endpoints for primary and replication MR analyses. Colocalization used IL6R-locus exposure and outcome regional statistics to evaluate shared causal-variant support. TCGA transcriptomic analyses used TCGA-COAD/READ expression and clinical data to evaluate tumour expression, reconstructed OS/PFI and immune context. Single-cell analysis used GSE146771 10x and Smart-seq2 data to localize cell-state and IL6-axis communication evidence. The support layer used co-expression, GO/KEGG/Reactome and drug-target annotation to contextualize immune/myeloid programmes.

Genetic exposure and outcome data for Mendelian randomization

Genetic instruments were prepared for IL6R and related inflammatory traits, including IL-6 and CRP where available, using harmonized summary-level genetic association files preserved in the project workflow. The primary exposure for manuscript interpretation was genetically proxied circulating IL6R protein level. CRC outcome analyses included a FinnGen colorectal cancer endpoint and two additional CRC-related GWAS endpoints used for replication and robustness checking. Harmonized output files included IL6R-to-FinnGen CRC, IL6R-to-IEU CRC and IL6R-to-EBI CRC analyses, allowing the primary and replication endpoints to be interpreted on a comparable analysis scale.

Genetic variants were retained as instrumental variables only when they satisfied the pre-specified relevance and harmonization criteria in the MR workflow. Palindromic and ambiguous variants were handled during harmonization, and alleles were aligned so that causal estimates corresponded to the same effect allele for exposure and outcome. The MR workflow followed the conceptual basis of instrumental-variable causal inference in genetic epidemiology and used multiple sensitivity analyses to reduce over-reliance on a single estimator [6,7].

Mendelian randomization and sensitivity analyses

The primary MR estimator was inverse-variance weighting (IVW). Weighted median, MR-Egger and weighted mode analyses were used as complementary estimators. Heterogeneity was assessed using Cochran's Q statistics, and horizontal pleiotropy was assessed using MR-Egger intercept tests. Leave-one-out analyses were used to evaluate whether the primary estimate was driven by a single variant. Results were reported as beta coefficients, odds ratios (ORs), 95% confidence intervals (CIs), P values and number of SNPs. MR reporting was aligned with STROBE-MR principles, emphasizing transparent reporting of data sources, assumptions, sensitivity analyses and limitations [8].

Colocalization at the IL6R locus

Colocalization was performed at the IL6R locus using exposure-region and outcome-region summary statistics preserved in the project release. The analysis used an approximate Bayesian framework to estimate posterior probabilities for five hypotheses: no association with either trait, association with the exposure only, association with the outcome only, association with both traits through different causal variants, and association with both traits through a shared causal variant. The key interpretive statistic was PP.H4, the posterior probability for a shared causal variant [9]. PP.H4 was interpreted conservatively; low PP.H4 was treated as evidence against strong shared-variant support rather than as proof that IL6R has no biological role in CRC.

TCGA-COAD/READ expression and clinical analyses

TCGA-COAD/READ transcriptomic and clinical analyses were performed using project-preserved TCGA data objects and output tables. IL6R expression was compared between adjacent normal and tumour tissues in combined COAD/READ and in COAD and READ strata. The project workflow used TCGA data structures generated through GDC/TCGA-compatible preprocessing, with TCGAbiolinks-style TCGA integration serving as the reference computational framework [10,11].

For clinical analysis, patient-level OS and PFI endpoints were reconstructed after an earlier survival-summary issue was identified. IL6R expression was matched to patient barcodes, and PFI information was supplemented from the TCGA Pan-Cancer Clinical Data Resource where required [12]. Continuous Cox models were used as the primary survival association analysis, while median-split Kaplan-Meier curves were generated as visual summaries. OS and PFI were assessed in the combined COAD/READ cohort and in COAD and READ strata. Results were interpreted as clinical association evidence, not as causal evidence.

Immune-infiltration analyses in TCGA

To contextualize IL6R expression in relation to tumour microenvironment composition, immune and stromal infiltration features were evaluated using MCP-counter and quanTIseq-style immune deconvolution outputs [13,14]. Spearman correlations were calculated between IL6R expression and immune-infiltration scores in COAD and READ. Immune-infiltration results were interpreted with emphasis on concordant patterns across myeloid, neutrophil, macrophage/monocyte, endothelial and T-cell-related signatures rather than single isolated cell types.

Single-cell RNA-seq and cell-cell communication analyses

Single-cell RNA-seq analyses were performed using the GSE146771 10x and Smart-seq2 datasets preserved in the project workflow. The analysis included quality-controlled expression matrices, dimensionality reduction, cell-state visualization, IL6-axis dot plots, module score calculation and cell-type-level summaries. Seurat-style single-cell workflows were used as the conceptual framework for normalization, clustering and visualization [15,16].

CellChat was used to infer ligand-receptor communication networks from single-cell expression data, with specific attention to IL6 -> IL6R_IL6ST signalling [17]. Smart-seq2 results were treated as the primary IL6-axis communication evidence because the Smart-seq2 workflow recovered IL6-axis communication edges, whereas the 10x workflow did not recover IL6-axis edges despite successful CellChat execution. The 10x result was interpreted as a sparse-platform comparison rather than proof of absent IL6-axis communication.

IL6R-centred co-expression, enrichment and drug-target support analyses

A support-layer analysis was performed to contextualize IL6R-associated transcriptomic biology in TCGA tumour samples. Spearman correlations between IL6R expression and genome-wide tumour expression were calculated, and an IL6R-associated support gene set was defined using positively correlated genes and manually curated immune, myeloid and inflammatory seed genes. A co-expression network was generated from the support gene set. Functional enrichment was performed using Gene Ontology Biological Process, KEGG and Reactome pathway resources through clusterProfiler-compatible workflows [18-20,34]. STRING and related network concepts were used only as background support for interpreting protein and pathway networks, not as direct evidence of causal biology [21].

Drug-target annotation was used to summarize candidate axis-level relationships involving IL6/IL6R/JAK/STAT and myeloid-associated targets. These relationships were interpreted as exploratory druggability context only. They were not interpreted as evidence that the listed agents are clinically validated CRC therapies or that IL6R blockade would necessarily benefit CRC patients.

Statistical analysis and reproducibility

All analyses were conducted using scripted R workflows and versioned output folders. Statistical tests were two-sided unless otherwise specified. Multiple-testing correction was applied where implemented in the workflow, including false discovery rate adjustment for tumour-normal expression and enrichment analyses. The project release snapshot contains result tables, figures, logs and manifest files for MR, colocalization, TCGA, single-cell and support-layer analyses. The manuscript reports versioned workflow outputs rather than manual recalculations.

External GEO validation

External GEO validation was added as an independent post hoc validation layer. GSE39582, GSE17536, GSE17537 and GSE14333 were queried using GEOquery and preserved in the project output directory. IL6R probes were identified from GPL570 platform annotations using the Gene Symbol field. For successfully parsed cohorts, sample-level IL6R expression was extracted from the selected primary IL6R probe and from all matched IL6R probes. Because no external cohort provided an automatically reliable tumour-normal split in the available metadata, external tumour-normal validation was not claimed. For GSE17536 and GSE17537, where outcome metadata were available, univariable Cox proportional-hazards models were fitted using z-scored IL6R primary-probe expression for OS, DFS and, where available, DSS. These analyses were treated as external validation of the prognostic interpretation rather than as a replacement for the primary TCGA analyses.

Results

Mendelian randomization did not support a significant causal effect of circulating IL6R on CRC risk

Two-sample MR was performed to evaluate whether genetically proxied circulating IL6R protein levels were associated with CRC risk. In the primary FinnGen CRC endpoint, the IVW estimate did not support a statistically significant association (beta = -0.0474, OR = 0.954, 95% CI 0.889-1.024, P = 0.190). Complementary weighted median, MR-Egger and weighted mode analyses were directionally and statistically consistent with a null or weak association rather than a robust causal effect. Replication endpoints also did not provide significant support: the IEU CRC endpoint yielded IVW P = 0.916, and the EBI Firth-corrected CRC endpoint yielded IVW P = 0.603. The MR summary with readable endpoint labels is shown in Figure 2.

Sensitivity analyses did not reveal strong evidence of violation of MR assumptions in the primary endpoint. In FinnGen CRC, heterogeneity and MR-Egger intercept tests did not indicate strong heterogeneity or directional pleiotropy, and leave-one-out analyses did not identify a single SNP that converted the overall estimate into a robust significant association. These findings indicate that the available genetic evidence did not support a strong causal effect of circulating IL6R protein levels on CRC risk.

Colocalization did not support a shared causal variant at the IL6R locus

Colocalization analysis was used to evaluate whether the IL6R protein-associated signal and the CRC risk-associated signal at the IL6R locus were compatible with a shared causal variant. The analysis included 1,062 SNPs and yielded PP.H4 = 0.048. This value was well below the threshold typically interpreted as strong support for colocalization. The posterior probability pattern was instead dominated by exposure-only evidence, indicating that the IL6R pQTL signal and the CRC association signal were not strongly supported as arising from the same causal variant. Therefore, the genetic evidence did not support a shared-variant mechanism linking circulating IL6R protein levels to CRC risk (Figure 3).

IL6R expression was markedly reduced in CRC tumours but was not a robust OS/PFI prognostic marker

In TCGA-COAD/READ, IL6R expression was significantly lower in tumour tissues than in adjacent normal tissues. In the combined cohort, median IL6R expression was 11.752 in normal tissues and 8.353 in tumours, with FDR = 1.09e-31. Similar tumour-normal decreases were observed in COAD and READ strata, supporting a consistent reduction of IL6R mRNA expression in CRC tumour tissue.

Because an earlier survival summary produced missing survival estimates, OS and PFI were reconstructed at the patient level. After this repair, IL6R expression was not significantly associated with OS or PFI. In the combined COAD/READ cohort, OS analysis included 551 patients and 121 events (log-rank P = 0.968; continuous Cox HR = 0.931, 95% CI 0.754-1.151, P = 0.511), while PFI analysis included 596 patients and 153 events (log-rank P = 0.546; continuous Cox HR = 1.052, 95% CI 0.871-1.271, P = 0.599). Similar null associations were observed in COAD and READ strata. These results indicate that IL6R is not a robust standalone prognostic marker in TCGA CRC, despite its strong tumour-normal expression difference (Figure 4).

IL6R expression was associated with immune-infiltrated and myeloid-enriched tumour microenvironments

IL6R expression showed strong associations with immune-infiltration features in TCGA CRC. MCP-counter analyses showed particularly strong correlations between IL6R expression and endothelial, neutrophil, macrophage/monocyte, myeloid dendritic-cell and T-cell-related signatures. In COAD, representative Spearman correlations included endothelial cell rho = 0.679, neutrophil rho = 0.676, macrophage/monocyte rho = 0.668 and T-cell rho = 0.653. In READ, representative correlations included neutrophil rho = 0.719, endothelial cell rho = 0.717 and macrophage/monocyte rho = 0.685. quanTIseq results provided additional immune-context support. These results suggest that IL6R expression in bulk CRC transcriptomes reflects immune and stromal composition, particularly myeloid and inflammatory tumour microenvironment features (Figure 5).

Single-cell analysis localized IL6-axis activity in the CRC tumour microenvironment

GSE146771 single-cell RNA-seq analysis included both 10x and Smart-seq2 data. The 10x dataset included 14,662 cells and 13,538 features, and the Smart-seq2 dataset included 10,322 cells and 15,179 features. UMAP visualizations, IL6-axis module-score summaries and cell-communication outputs were generated for both platforms.

CellChat analysis supported platform-dependent IL6-axis communication. In Smart-seq2, CellChat detected 35,283 total communication edges and 32 IL6-axis edges, including IL6 -> IL6R_IL6ST interactions. In contrast, the 10x analysis successfully ran CellChat but did not recover IL6-axis edges. The Smart-seq2 results were therefore treated as the primary evidence for IL6-axis communication, while the 10x results were interpreted as a sparse-platform comparison. Together, these findings support the presence of IL6-axis activity in specific CRC tumour microenvironment contexts (Figure 6).

IL6R-centred support-layer analyses reinforced immune, myeloid and inflammatory biology

To further characterize the transcriptomic context of IL6R in CRC, an IL6R-centred tumour co-expression network was constructed from TCGA-COAD/READ tumour samples. The resulting support-layer network contained 109 support genes, 109 nodes and 4,933 co-expression edges. Functional enrichment analysis identified 895 Gene Ontology Biological Process records, 54 KEGG records and 227 Reactome records. The strongest terms converged on immune-response activation, leukocyte activation, chemokine and cytokine signalling, myeloid-cell activation and inflammatory response programmes.

Exploratory drug-target annotation was used to map IL6/IL6R/JAK/STAT and myeloid-associated nodes to candidate drug-target relationships. Anti-IL6R agents, anti-IL6 therapy, JAK inhibitors, a STAT3-pathway inhibitor and a CSF1R inhibitor were represented as exploratory axis-level candidates. These results should be interpreted as hypothesis-generating support for IL6R-associated immune/myeloid biology, rather than as evidence of clinically validated CRC therapy (Figure 7).

External GEO validation supported the non-prognostic interpretation of IL6R expression

External GEO validation confirmed that IL6R expression could be extracted from three independent CRC expression cohorts. Three GPL570 IL6R probes were identified in GSE17536, GSE17537 and GSE14333, yielding sample-level IL6R expression for 522 external samples (GSE17536 n = 177, GSE17537 n = 55 and GSE14333 n = 290). GSE39582 could not be parsed in the current automated run because the GEO/GPL download timed out and is therefore retained as a planned additional validation cohort rather than being used for inference. The selected primary probes were 205945_at in GSE17536, 226333_at in GSE17537 and 217489_s_at in GSE14333. Univariable external survival checks did not support a robust prognostic association. In GSE17536, z-scored IL6R expression was not associated with OS (HR = 0.897, 95% CI 0.708-1.136, P = 0.366), DFS (HR = 1.003, 95% CI 0.722-1.392, P = 0.987) or DSS (HR = 0.922, 95% CI 0.703-1.210, P = 0.560). In GSE17537, IL6R expression was also not significantly associated with OS (HR = 1.327, 95% CI 0.864-2.037, P = 0.196) or DFS (HR = 0.563, 95% CI 0.204-1.554, P = 0.267). These external results were consistent with the repaired TCGA OS/PFI analysis and support the interpretation that IL6R is not a robust standalone prognostic marker in CRC. The GEO validation did not provide a reliable independent tumour-normal comparison because no cohort had a sufficiently unambiguous automated tumour-normal split in the available phenotype metadata (Supplementary Figure S1; Supplementary Tables S1-S7).

Discussion

This integrative study separated genetic-causality evidence from transcriptomic and single-cell evidence to clarify the role of IL6R in CRC. The main finding is deliberately conservative: genetically proxied circulating IL6R protein levels were not significantly associated with CRC risk in MR analyses, and colocalization did not support a shared causal variant at the IL6R locus. These results argue against a strong germline causal interpretation for circulating IL6R in CRC risk under the available instruments and outcome definitions. At the same time, TCGA and single-cell analyses consistently linked IL6R to immune-infiltrated, myeloid-enriched and inflammatory CRC tumour microenvironments. This distinction is central to the manuscript: IL6R should not be framed as a proven causal CRC-risk driver, but as an immune-context marker and potential signalling node within tumour-microenvironment biology.

The negative MR and colocalization results are biologically informative. MR is useful for testing whether lifelong genetically proxied differences in an exposure influence disease risk, but it does not capture every biological context of a molecule [6,7]. In this study, the exposure represented circulating IL6R protein levels, whereas tumour biology may depend on membrane-bound receptor expression, soluble receptor dynamics, local IL-6 production, IL6ST availability, stromal interactions and cell-type-specific signalling. Colocalization further indicated that the IL6R pQTL signal and CRC risk signal were not strongly supported as the same causal variant, making a direct shared-variant claim inappropriate [9]. This combination of results supports a cautious interpretation: the available genetic evidence does not establish circulating IL6R as a causal determinant of CRC risk.

The TCGA tumour-normal result provides a different layer of evidence. IL6R mRNA expression was markedly lower in CRC tumours than in adjacent normal tissues, yet this expression difference did not translate into a robust OS/PFI prognostic association after patient-level survival reconstruction. This pattern suggests that IL6R expression may reflect tissue composition or immune/stromal architecture rather than tumour-cell-intrinsic aggressiveness. The use of TCGA-CDR-supported PFI reconstruction was important because clinical endpoint quality strongly affects prognostic interpretation in TCGA analyses [12]. The repaired survival findings therefore refine the claim: IL6R is not a strong standalone survival marker in this cohort, even though it remains biologically meaningful in tumour immune-context analysis.

The immune-infiltration results provide the strongest tissue-level explanation for this pattern. IL6R expression correlated with endothelial, neutrophil, macrophage/monocyte, myeloid dendritic-cell and T-cell-related signatures, pointing to an immune-infiltrated and myeloid-enriched tumour microenvironment. This is consistent with the broader biology of the IL-6/IL6R/JAK/STAT3 axis, which can shape cytokine signalling, immune-cell recruitment, stromal activation, angiogenesis and inflammatory feedback [3,4,29,30]. It also explains why IL6R may be transcriptomically informative despite negative MR, colocalization and survival-marker results: bulk IL6R expression may partly represent immune and stromal cellular composition rather than an isolated tumour-cell oncogenic dependency.

Single-cell results strengthened the cellular-context interpretation. GSE146771 analysis showed that IL6-axis expression and module scores could be localized across tumour microenvironment compartments. Smart-seq2 CellChat analysis detected IL6 -> IL6R_IL6ST interactions, supporting ligand-receptor communication through the canonical IL6R/IL6ST receptor complex [17]. The absence of recovered IL6-axis edges in 10x data should be interpreted carefully because single-cell platforms differ in sparsity, transcript capture and sensitivity [15,16]. For this reason, Smart-seq2 was used as the primary CellChat communication evidence, while 10x was treated as a conservative platform comparison.

The support-layer network, enrichment and drug-target analyses further contextualized IL6R-associated biology. The IL6R-centred co-expression network and GO/KEGG/Reactome enrichment outputs converged on immune activation, cytokine-mediated signalling, myeloid leukocyte activation, chemotaxis and inflammatory pathways. These results align with the TCGA immune-infiltration and single-cell communication findings and support a coherent tumour-microenvironment model [18-21]. The drug-target network placed IL6R within an IL6/IL6R/JAK/STAT and myeloid-associated axis. However, these annotations are exploratory. They are best viewed as druggability context for an immune/myeloid axis, not as evidence that the listed agents are established CRC therapies or that targeting IL6R would necessarily improve CRC outcomes.

The external GEO validation layer reduced an important limitation of the original computational study. IL6R expression was successfully extracted from three independent CRC expression cohorts, and univariable survival checks in GSE17536 and GSE17537 did not show robust OS, DFS or DSS associations. This finding is concordant with the repaired TCGA OS/PFI analysis and strengthens the conclusion that IL6R expression is unlikely to function as a standalone prognostic marker. However, the external validation remains incomplete: GSE39582 could not be parsed in the current run because of a download timeout, and the extracted GEO cohorts did not support a reliable automated tumour-normal split. Therefore, the external validation should be interpreted as a prognostic consistency check rather than a full replication of the tumour-normal expression analysis or of the immune-microenvironment findings.

Several limitations should be acknowledged. First, MR estimates depend on the strength, specificity and biological interpretation of available IL6R protein instruments. Circulating IL6R may not fully capture tissue-specific receptor biology, soluble versus membrane-bound receptor effects, or local IL6R/IL6ST signalling. Second, colocalization depends on locus coverage, ancestry compatibility, phenotype definitions and the single-causal-variant assumptions of the approximate Bayes factor framework [9]. Third, TCGA analyses are observational and cannot establish causal direction. Bulk immune-infiltration scores are computational estimates and may be influenced by tumour purity, stromal content and compositional effects. Fourth, single-cell analyses are affected by sample composition, platform sparsity, cell-state annotation and ligand-receptor inference assumptions. Fifth, the drug-target analysis is curated and exploratory rather than pharmacological validation. Finally, the present study requires public deposition of processed outputs and analysis code before final submission to maximize reproducibility.

The independent GEO validation also has limitations. It was based on public microarray cohorts and univariable survival models, and one planned large cohort (GSE39582) remained unavailable in the current automated run because of a GEO/GPL download timeout. Moreover, the GEO metadata did not support a reliable automated tumour-normal split, so the TCGA tumour-normal expression finding still requires additional external expression validation with manually curated phenotype annotation or another independent dataset.

Overall, the findings support a revised model of IL6R biology in CRC. IL6R is not supported by the present genetic evidence as a causal CRC risk factor, nor by repaired TCGA survival analyses as a standalone prognostic marker. Instead, IL6R appears to mark or participate in immune-infiltrated, myeloid-enriched and inflammatory CRC tumour microenvironments. Future studies should test whether IL6R-associated immune states predict immunotherapy response, inflammatory molecular subtypes, spatial immune organization, stromal niches or treatment-specific vulnerabilities in independent CRC cohorts and experimental systems.

Conclusions

This integrative analysis did not support a strong genetic causal or colocalized effect of circulating IL6R protein levels on CRC risk. Repaired TCGA survival analyses and external GEO survival checks also did not support IL6R as a robust standalone prognostic marker. However, TCGA immune-infiltration, single-cell and support-layer analyses consistently linked IL6R to an immune-, myeloid- and inflammatory-associated CRC tumour microenvironment. IL6R should therefore be interpreted primarily as a context-dependent microenvironmental marker and pathway component rather than as a genetically supported causal risk factor or isolated prognostic biomarker in CRC.

Abbreviations

CRC: colorectal cancer; MR: Mendelian randomization; IVW: inverse-variance weighted; OR: odds ratio; CI: confidence interval; TCGA: The Cancer Genome Atlas; COAD: colon adenocarcinoma; READ: rectum adenocarcinoma; OS: overall survival; PFI: progression-free interval; scRNA-seq: single-cell RNA sequencing; TME: tumour microenvironment; GO: Gene Ontology; KEGG: Kyoto Encyclopedia of Genes and Genomes; PP.H4: posterior probability of a shared causal variant.

Declarations

Ethics approval and consent to participate

This study used publicly available, de-identified summary-level GWAS data, TCGA-derived bulk transcriptomic and clinical data, TCGA-CDR survival annotations and public single-cell RNA-seq data. No new human participants, biospecimens or animal experiments were involved. Additional ethics approval and consent to participate were therefore not required for this secondary analysis of public resources.

Consent for publication

Not applicable.

Availability of data and materials

All analyses used publicly available or project-processed derivatives of public resources. GWAS summary statistics were obtained from FinnGen, IEU/OpenGWAS and EBI/GWAS Catalog sources as specified in the Methods. TCGA analyses used publicly available TCGA-COAD/READ and TCGA-CDR resources. Single-cell analyses used the public GSE146771 dataset. External validation used public GEO datasets GSE17536, GSE17537, GSE14333 and the attempted GSE39582 query. Project-processed tables, figure source data and GEO validation outputs are organized in the accompanying supplementary data package; a permanent repository URL or DOI remains to be added before final submission: [[INSERT_REPOSITORY_URL_OR_DOI]].

Competing interests

The author declares no competing interests.

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Authors’ contributions

ZC conceived and designed the study, performed the computational analyses, interpreted the results, prepared the figures and tables, drafted the manuscript and approved the submitted version.

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Figure legends

Figure 1. Study workflow and integrated evidence chain. The workflow summarizes the integrated evidence-chain strategy used to evaluate IL6R in colorectal cancer, including pQTL-based Mendelian randomization, IL6R-locus colocalization, TCGA COAD/READ transcriptomic and reconstructed survival analyses, GSE146771 single-cell analysis, and support-layer network, enrichment and drug-target analyses. The final conclusion emphasizes that MR and colocalization did not support strong genetic causal or shared-variant evidence, whereas downstream transcriptomic and cellular analyses linked IL6R to an immune-, myeloid- and inflammatory-associated tumour microenvironment.

Figure 2. Mendelian randomization analyses of genetically proxied circulating IL6R and colorectal cancer risk. The figure summarizes method-specific MR estimates, statistical strength and directionality across the primary FinnGen CRC endpoint and replication CRC GWAS endpoints using readable endpoint labels. The estimates do not provide strong evidence that genetically proxied circulating IL6R has a robust causal effect on colorectal cancer risk.

Figure 3. Colocalization analysis at the IL6R locus. Panel A shows posterior probabilities for coloc hypotheses H0-H4. Panel B provides a compact posterior-evidence summary, and Panel C summarizes the main colocalization conclusion. PP.H4 was approximately 0.048, indicating limited evidence that circulating IL6R protein and colorectal cancer risk share a common causal variant at the IL6R locus.

Figure 4. TCGA expression and reconstructed OS/PFI survival analyses. The figure summarizes TCGA patient-level IL6R expression distributions and reconstructed OS/PFI continuous Cox estimates. These analyses support the interpretation that IL6R is not a robust standalone prognostic marker in TCGA colorectal cancer despite its strong tumour-normal expression difference and immune-microenvironment associations.

Figure 5. IL6R-associated immune infiltration patterns in TCGA colorectal cancer. The traceable immune-infiltration figure summarizes correlations between IL6R expression and immune/stromal infiltration estimates across deconvolution methods. Dot colour represents Spearman rho and dot size represents statistical strength using a capped -log10(FDR) visual scale; full rho and FDR values are retained in the source tables. The figure emphasizes myeloid, endothelial, neutrophil, macrophage/monocyte and inflammatory tumour-microenvironment features.

Figure 6. Single-cell evidence for IL6-axis activity in GSE146771. The single-cell redraw summarizes 10x and Smart-seq2 cell-state structure, IL6-axis module evidence and CellChat-supported communication evidence from traceable source tables. Smart-seq2 provides the clearer IL6-axis communication signal, while the 10x dataset is retained as a sparsity-sensitive comparator. The figure is interpreted as transcriptomic and cell-communication context rather than as independent causal evidence.

Figure 7. IL6R-associated support-layer network, enrichment and drug-target evidence. The label-safe figure combines an IL6R-centred support network with immune/inflammatory pathway enrichment and exploratory drug-target support across IL6/IL6R/JAK/STAT and myeloid-associated axes. IL6R-connected edges were

retained in the final network rendering. Drug-target annotations are hypothesis-generating and are not interpreted as evidence of clinically validated colorectal cancer therapy.